GPU Optimisation of an Image Classification Pipeline with TensorFlow and Keras.

Introduction

This assignment investigated the effect of GPU-aware optimisation techniques on a TensorFlow image classification pipeline using the CIFAR-10 dataset. A baseline convolutional neural network (CNN) was developed and subsequently optimised through improvements to both the data pipeline and model architecture. Performance was evaluated using loss curves, accuracy curves, prediction examples and a confusion matrix.

Baseline Model Performance

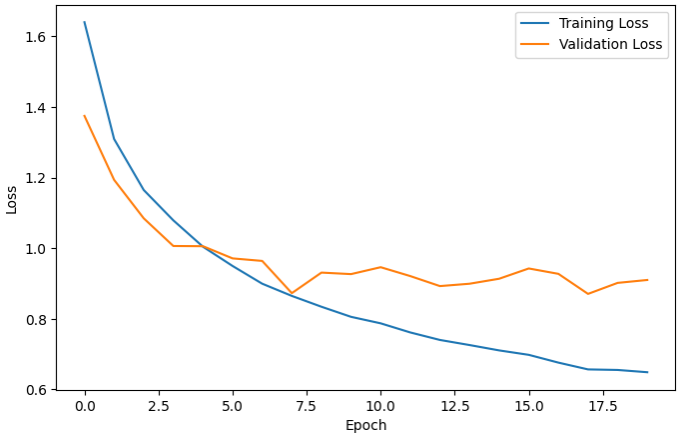


Figure 1: Baseline Training and Validation Loss

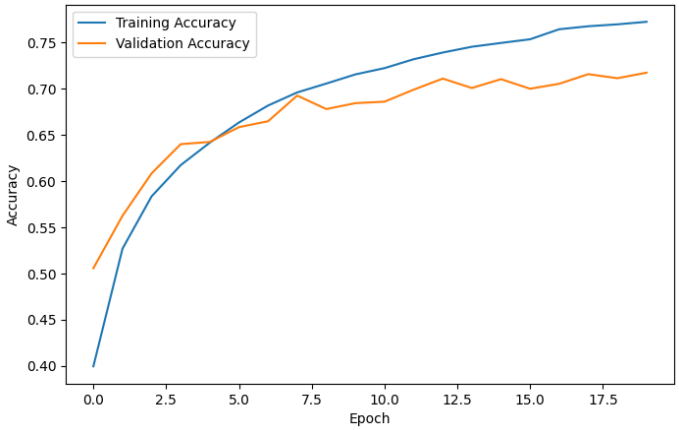


Figure 2: Baseline Training and Validation Accuracy



Figure 3: Baseline Prediction Examples

The baseline CNN was able to learn some good feature representations for the images (validation accuracy increased to about 72%). Loss during training decreased continuously from 1.64 to about 0.65. Validation loss also decreased to about 0.91. At about epoch 8, however, training loss continued to decrease whereas validation loss remained relatively constant. This would suggest that the CNN started to overfit the training data. The example predictions show that the CNN was able to correctly classify a number of classes such as horses, frogs, trucks, and automobiles. There were

examples where an image of an airplane was misclassified as a ship. One possible reason for this could be the similarity in colour distributions and background features between these two types of objects.

Data Pipeline and Model Optimisation

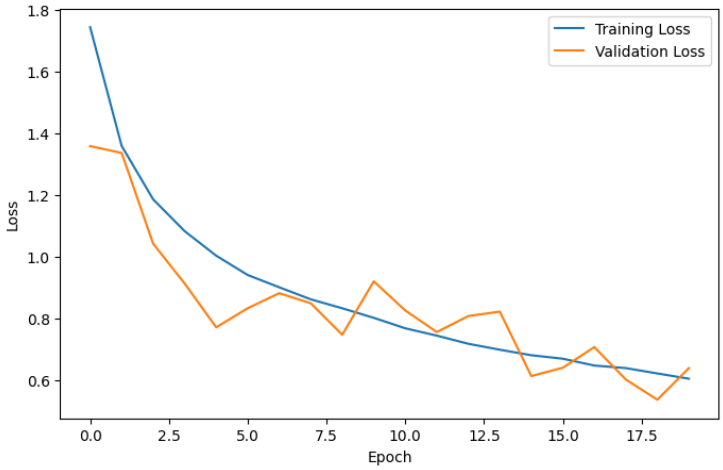


Figure 4: Optimised Training and Validation Loss

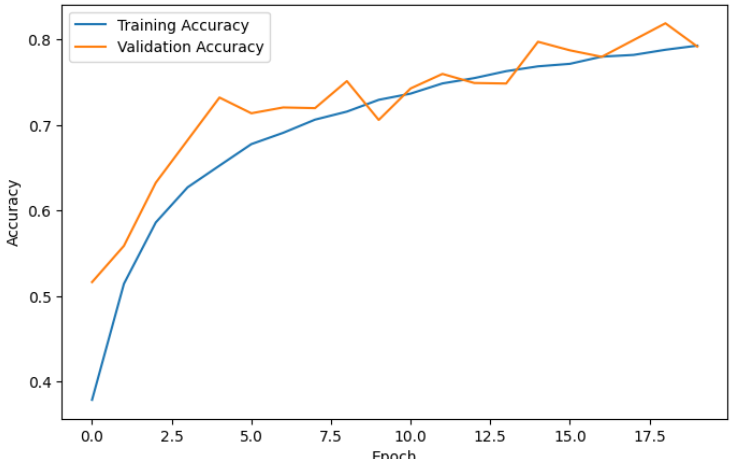
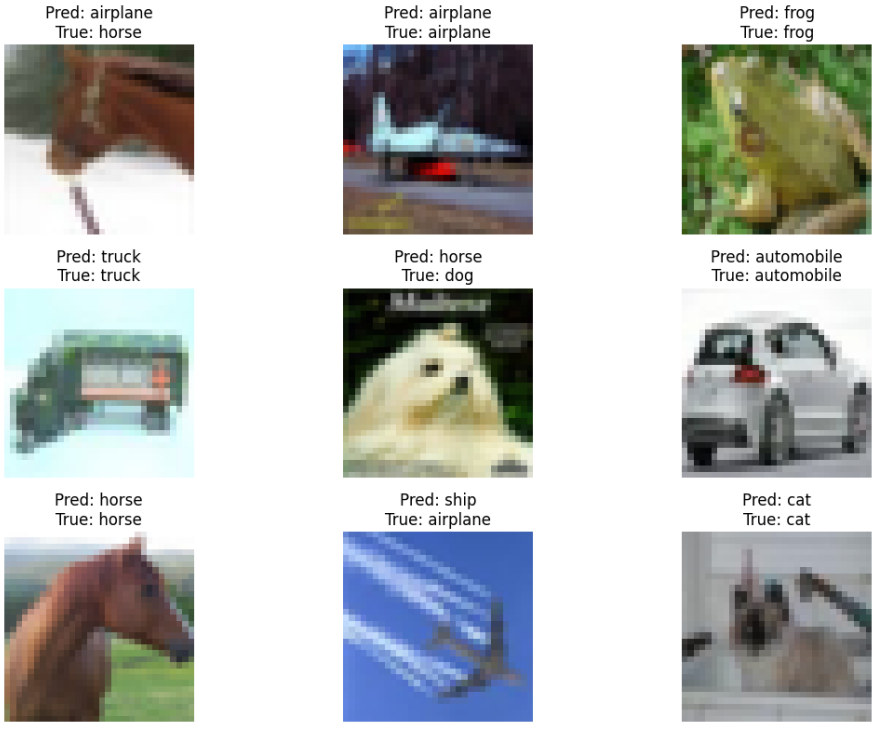


Figure 5: Optimised Training and Validation Accuracy

Optimisations performed on the TensorFlow data pipeline included data loading caching, mapping in parallel and prefetching. Such strategies minimised the bottlenecks related to the loading of data and maximised the efficiency of the GPU during training. The CNN was upgraded via the inclusion of the Batch Normalisation and Dropout operations.

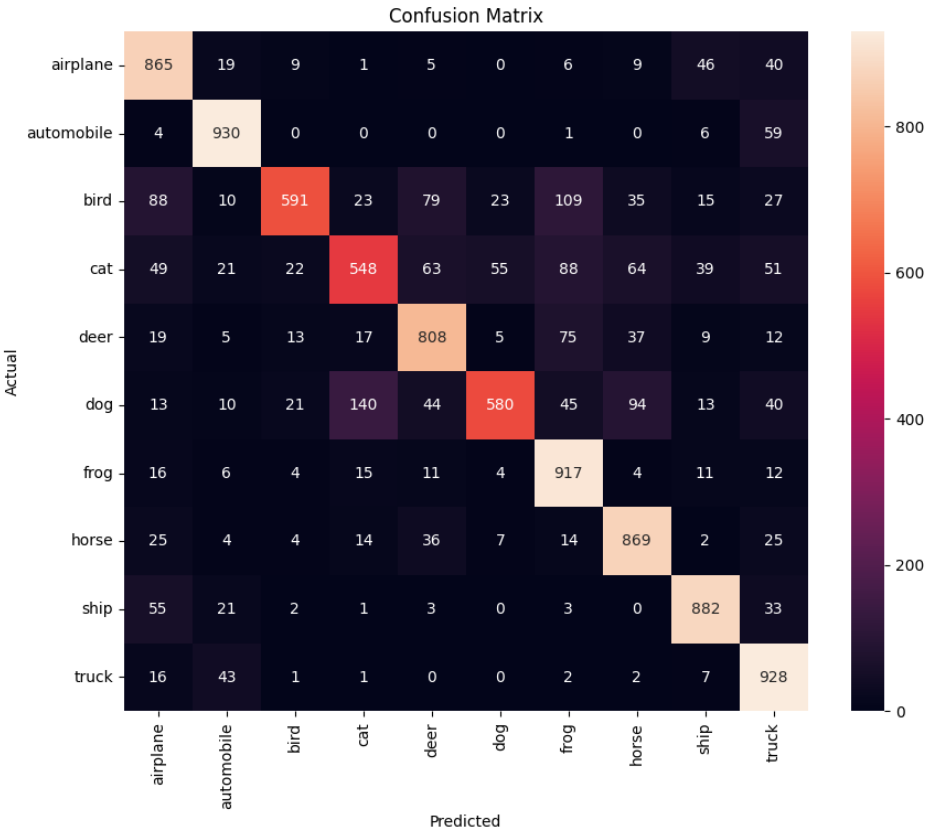


Optimisations in the model showed significant improvement compared to the baseline implementation. The accuracy for validation improved by an increment of 79.2%, from an estimated 71.8%. In addition, the loss value reduced from 0.91 to 0.63. Also, the performance discrepancy between training and validation became very low.

The sample outputs show that the optimised model accurately categorised most of the image classes, only having a few remaining errors.

Figure 6: Optimised Prediction Examples

Evaluation of the Optimised Model



According to the confusion matrix, the vast majority of the predictions are made diagonally, showing how well the model classifies its data. The top-performing classes were those of automobile, truck, frog, ship, and horse. These classes managed to obtain high levels of accurate predictions. The hardest classes to classify, however, were those of bird, cat, and dog. Dogs tended to be confused with cats and horses, while birds were occasionally classified as frogs or deer. Such mistakes are predictable given that CIFAR-10 images are 32×32 in size and animals have similar textures and shapes.The confusion matrix proves that the optimised model learned to distinguish better between classes.

Figure 6: Optimised Prediction Examples

Conclusion

Optimisation of TensorFlow pipeline and CNN model showed impressive performance in classification. The validation accuracy was raised from around 71.8% to 79.2%. Also, the validation loss was lowered drastically. Overfitting was avoided by using Batch Normalisation and Dropout effectively.

From the above findings, it is clear that the use of effective techniques in data processing in GPU with proper regularisation leads to optimal performance in deep learning classification tasks.